

Consumption of Instagram and Twitter APIs and the use of web scraping for the detection of pet adopter profiles

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Abstract

This research addresses the growing abandonment of pets in the city of Morelia and proposes a system that integrates the consumption of Instagram and X (Twitter) APIs along with web scraping techniques to extract data from online profiles and animal cruelty cases from local websites. A machine learning model capable of predicting the suitability of potential adopters is proposed, contextualized by a georeferenced "heat map" of the city. After facing limitations with API access, a prototype heat map was developed that identifies areas with a higher incidence of failed adoptions. It is concluded that these tools can optimize adoption processes, and collaboration with shelters is recommended to improve the quality and quantity of data.

Keywords:

- API consumption
- Web scraping
- Pet adoption
- Machine learning
- Social media profiling
- Animal welfare
- Adoption heat map
- Geospatial analysis

Introduction

Background

Pet abandonment is a phenomenon of increasing concern in various societies, reflecting not only an animal welfare problem but also social and psychological issues that affect communities. In Morelia, during April and May 2024, pet abandonment increased by 40%.

More recently, in 2025, the "Rescatistas Independientes" collective reported a worrying increase of up to 50% in abandoned pet reports in Morelia, especially during holidays. A primary cause is the lack of long-term commitment required for pet ownership.

In 2024, Morelia recorded up to 10,000 sterilized dogs, yet an estimated 70,000 remain on the streets. In addition to abandonment, there is the problem of pets being adopted for purposes of abuse or other objectives that end in the animal's death or permanent harm. In Morelia, 30 cases of animal cruelty were reported in 2023, and these are only registered cases. This cruelty can be attributed to a lack of education on responsible pet care, socioeconomic problems, and the normalization of violence in some cultures. The COVID-19 pandemic worsened this situation in Mexico, with a reported 15% increase in abandonment and abuse cases due to changes in families' personal and economic circumstances.

Context

Despite technological advances, specific applications for effectively tracking animal welfare or facilitating the reporting of abuse have not been developed. Existing applications for pet registration do not conduct detailed follow-ups on the actual situation of each animal. To date, there is no application, program, or tool specifically focused on efficiently addressing the issue of pet adoption.

Problem Statement

While applications have been proposed to monitor and properly record the pet adoption process, they have not had a significant impact. This demonstrates a lack of technological tools to facilitate the evaluation and monitoring of potential adopters, which contributes to the persistence of problems like abandonment and abuse. Therefore, there is a need to implement a machine learning model to determine the suitability of adopters and ensure the well-being of adopted pets. The research also aims to identify patterns to establish a "heat map" that can correlate reports of abuse with adopters who have already undergone a suitability process, seeking to identify patterns or correlations in such cases.

Objectives

General Objectives

- Develop a machine learning model to determine the suitability of potential pet adopters, thereby improving adoption processes and promoting animal welfare.
- Gather and analyze data from the online profiles of individuals interested in adopting to identify patterns and characteristics that indicate whether an applicant meets the minimum requirements to be considered a good adopter.

- Obtain data from web pages that address animal cruelty to understand the context of adoptions in the city and the associated problems.

Specific Objectives

1. Implement a machine learning model to analyze data related to pet adoption.
2. Implement techniques such as web scraping and technologies like APIs for data collection.
3. Evaluate the effectiveness of using machine learning techniques to reduce bias in adopter selection.
4. Develop a set of objective indicators that can be used by pet adoption organizations.
5. Assess the effectiveness of exclusively using online data for filtering, without resorting to traditional data collection methods like interviews

Methodology

Universe and Sample Selection

The universe includes all available profiles on selected social networks, as well as all types of web pages (blogs, news sites, etc.) that discuss animal cruelty in Morelia, Michoacán.

Inclusion Criteria for Social Media Profiles:

- Profiles older than one year.
- Profiles with a verifiable location in Morelia, Michoacán.
- Public profiles with explicit consent for data use.

Inclusion Criteria for Websites:

- Official, governmental, or trusted media sources.
- Sites focused on animal cruelty specifically in Morelia.
- Pages without legal restrictions on the use of their information.

A sample size of approximately **384 people** was calculated, resulting in **768 social media profiles** to be analyzed (assuming two profiles per person). Similarly, **384 articles** from websites will be collected.

Data Collection Instruments

The plan involved using web scraping and APIs from Twitter (X) and Instagram.

- **Web Scraping:** Python libraries such as BeautifulSoup, Scrapy, and Selenium were to be used to extract information from forums, blogs, and websites related to animal cruelty. Selenium was chosen for its ability to handle dynamic content.
- **Twitter (X) API:** The intention was to obtain tweets from specific users, search by keywords, analyze interactions, and extract profile information. However, the cost of the API was prohibitive for this research.
- **Instagram API:** The goal was to obtain profile information, posts, and follower data. However, Instagram's terms of use and the lack of an official API prevented scraping with third-party tools. The necessary permissions were requested but ultimately denied by Meta's Data Protection Officer.

Due to these limitations, the data collection was restricted to web scraping of articles and a small, pre-existing dataset of good and bad adopters.

Results and Discussion

Heat Map of Adoptions

Given the constraints, the project pivoted to creating a heat map of Morelia using the collected data. The data was cleaned and geocoded to map the locations of good and bad adoption cases. Using Python, HTML, and JavaScript, an interactive heat map was developed that reads data from an Excel file and allows for the addition of new cases and data export.

The heat map revealed that the majority of bad adopter cases are concentrated on the outskirts of the city, in areas such as Villas del Pedregal, La Aldea, and Villas del Oriente. This finding aligns with local socioeconomic studies, suggesting that economic factors may have a greater influence than education on the ability to provide adequate care for a pet.

Discussion

The initial hypothesis—that a technological tool could help analyze data to improve the adoption process—was validated through the development of the heat map. The denial of access to Instagram's API and the high cost of X's API were significant limitations that radically altered the research's scope but provided valuable insights into the complexities of handling personal data. A preliminary machine learning model was developed in collaboration with colleagues, but due to a lack of sufficient data, its reliability is low and its use is not yet recommended.

Conclusions

The primary goal of developing a reliable machine learning model was not fully achieved due to the inability to access social media data via APIs, which forced a change in the data collection strategy. However, the research successfully pivoted to create a valuable tool—the heat map—which can help identify areas for targeted adoption and awareness campaigns.

A major contribution of this investigation is that it serves as a guide for future projects by thoroughly exploring the possibilities and restrictions of obtaining and using social media data. It also highlights the versatility of programming languages like Python, which was used to create the heat map after the initial machine learning plan was hindered.

A significant limitation was the lack of precise location information in the scraped online articles about animal cruelty, which made accurate mapping difficult. The main recommendation is to collaborate directly with shelters and adoption agencies to obtain higher quality data and to retrain the machine learning model as more data becomes available.

Overall, the research objective was met under the circumstances. It provides a foundation for future researchers, offering a mapping of high-risk areas in the city and underscoring the need for standardized data collection in the field of animal welfare. This work has the potential to help predict future cases of abuse based on geographic location and to allocate resources more effectively for sterilization and awareness programs.